

PERFUME E-COMMERCE

Istanbul Aydin University

EDA ANALYSIS

Exploratory Data Analysis of Perfume E-Commerce Dataset



INTRODUCTION

This project focuses on:

- Understanding the general structure of the dataset
- Analyzing perfume brands and product listings
- Detecting missing values and data quality issues
- Examining price distribution and outliers
- Comparing prices by gender category
- Observing listing trends over time
- Exploring relationships between numerical features



DATASET OVERVIEW

Perfume E-Commerce Dataset

Perfume E-Commerce Product Dataset for Analysis



Source: Kaggle

Data source: eBay perfume product listings

Number of samples: 2,000 perfume listings

Number of files: 2 CSV files: men's perfumes and women's perfumes

Number of features: 10 original features

Dataset type: E-commerce / product listing dataset

Target variable: There is no fixed target variable directly given in the dataset.

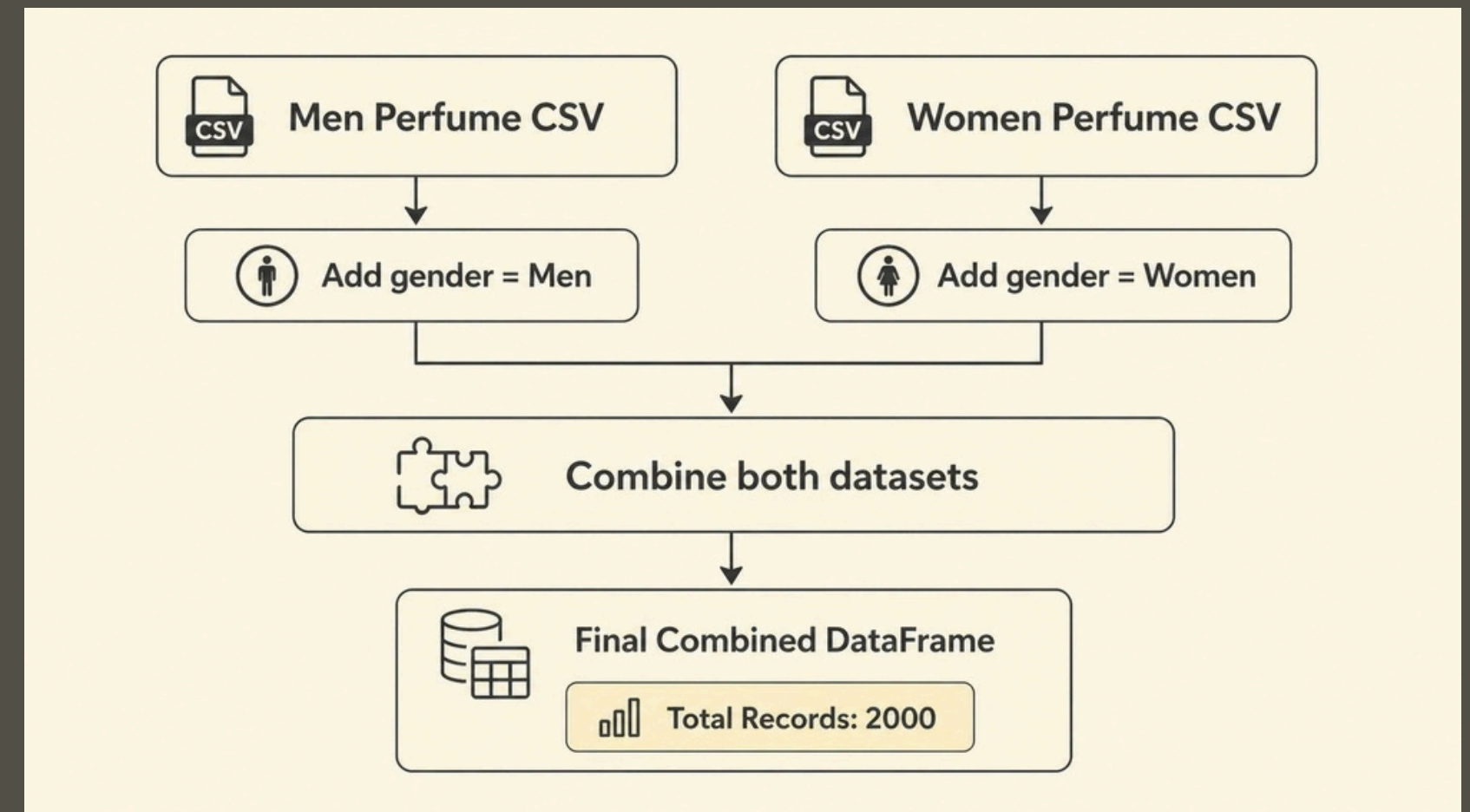
Problem type: Mainly Exploratory Data Analysis. It can also be used for regression if we choose price as the target variable.



DATA PREPARATION

Data preparation steps:

1. Loaded the men's perfume CSV file
2. Loaded the women's perfume CSV file
3. Added a new column called "gender"
4. Combined both files into one DataFrame
5. Checked rows, columns, data types, and missing values



```
import pandas as pd
```

```
# Load CSV files
```

```
men = pd.read_csv("ebay_mens_perfume.csv")
```

```
women = pd.read_csv("ebay_womens_perfume.csv")
```

```
# Add gender column
```

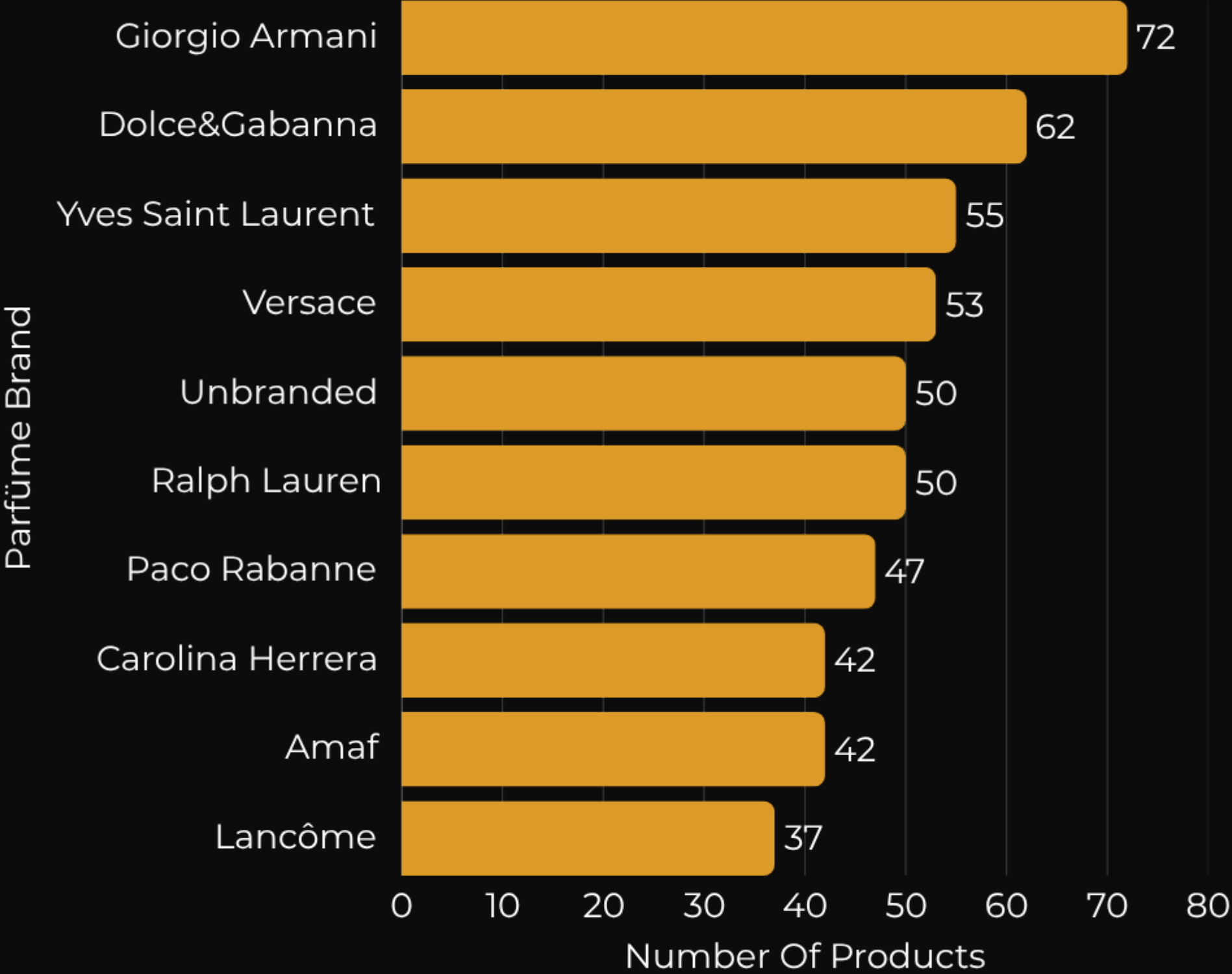
```
men["gender"] = "Men"
```

```
women["gender"] = "Women"
```

```
# Combine both datasets
```

```
df = pd.concat([men, women], ignore_index=True)
```

BRAND ANALYSIS



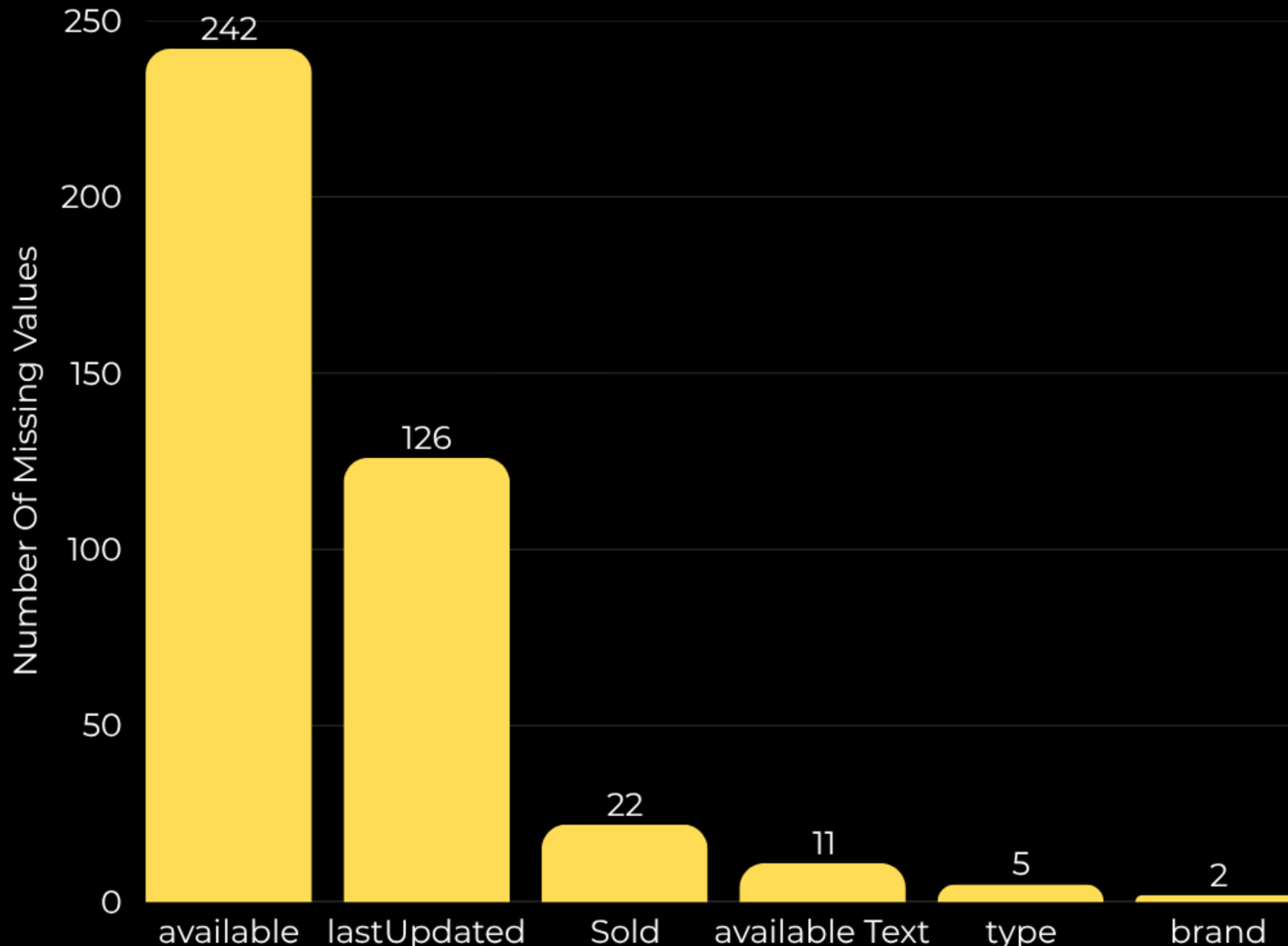
- **Technique:** Top 10 Bar Chart / Categorical Frequency Analysis
- **Variable Type:** Univariate - 1 Categorical Feature
- **Feature Used:** brand

Key Insights:

- Giorgio Armani is the most frequent brand with 72 products.
- Dolce & Gabbana and Yves Saint Laurent follow with 62 and 55 products.
- The dataset is dominated by well-known luxury perfume brands.
- Brand distribution is not equal; some brands appear much more often than others.

MISSING VALUE ANALYSIS

Missing Value Analysis



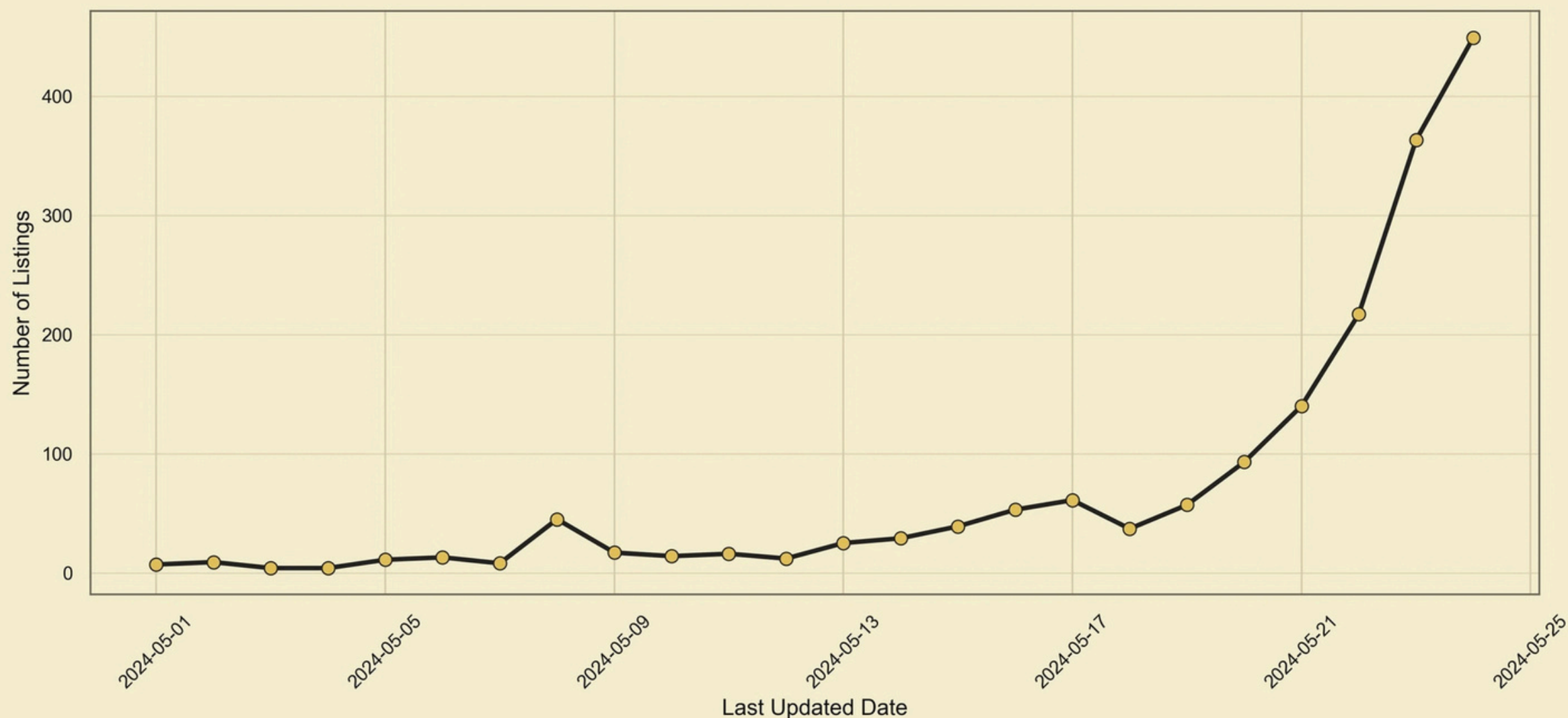
- **Technique:** Missing Value Bar Chart / Data Quality Analysis
- **Variable Type:** Multivariate - Multiple Dataset Columns
- **Features Used:** available, lastUpdated, sold, availableText, type, brand

Key Insights:

- **available** has the highest number of missing values with 242 missing records.
- **lastUpdated** has the second highest missing count with 126 missing records.
- **sold, availableText, type, and brand** have fewer missing values.
- Missing values are mostly related to availability and update information.
- These columns may need cleaning or imputation before further analysis.

LISTING TREND ANALYSIS

Listing Count Trend by Last Updated Date



- **Technique:** Line Chart / Time-Based Trend Analysis
- **Variable Type:** Bivariate - 1 Date Feature + 1 Numeric Count
- **Features Used:** lastUpdated, listing count

Key Insights:

- At the beginning of May 2024, the number of listings is low and relatively stable.
- After May 20, 2024, the listing count increases sharply.
- The highest listing count appears near the end of the observed period.
- This may show that many perfume listings were updated or added toward the end of May.
- The chart reveals a strong upward trend in listing activity.



CORRELATION HEATMAP ANALYSIS

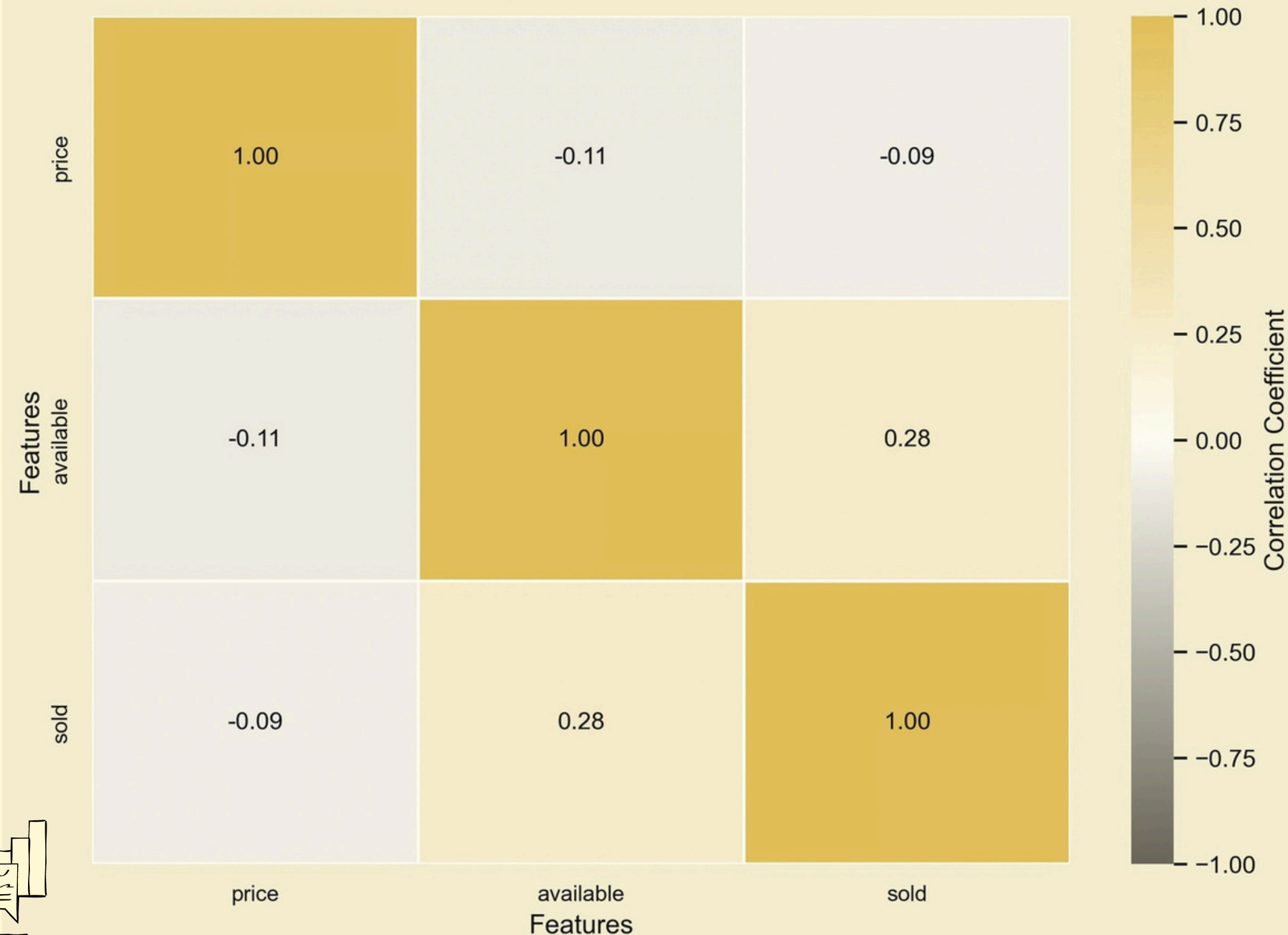
- **Technique:** Correlation Heatmap / Relationship Analysis
- **Variable Type:** Multivariate - Multiple Numerical Features
- **Features Used:** price, available, sold

Key Insights:

- **available** and **sold** have a weak positive correlation of 0.28.
- This means products with higher availability may have slightly higher sales.
- **price** and **available** have a very weak negative correlation of -0.11.
- **price** and **sold** also have a very weak negative correlation of -0.09.
- Overall, **price** does not show a strong linear relationship with sales or availability.



Correlation Heatmap of Numeric Listing Features

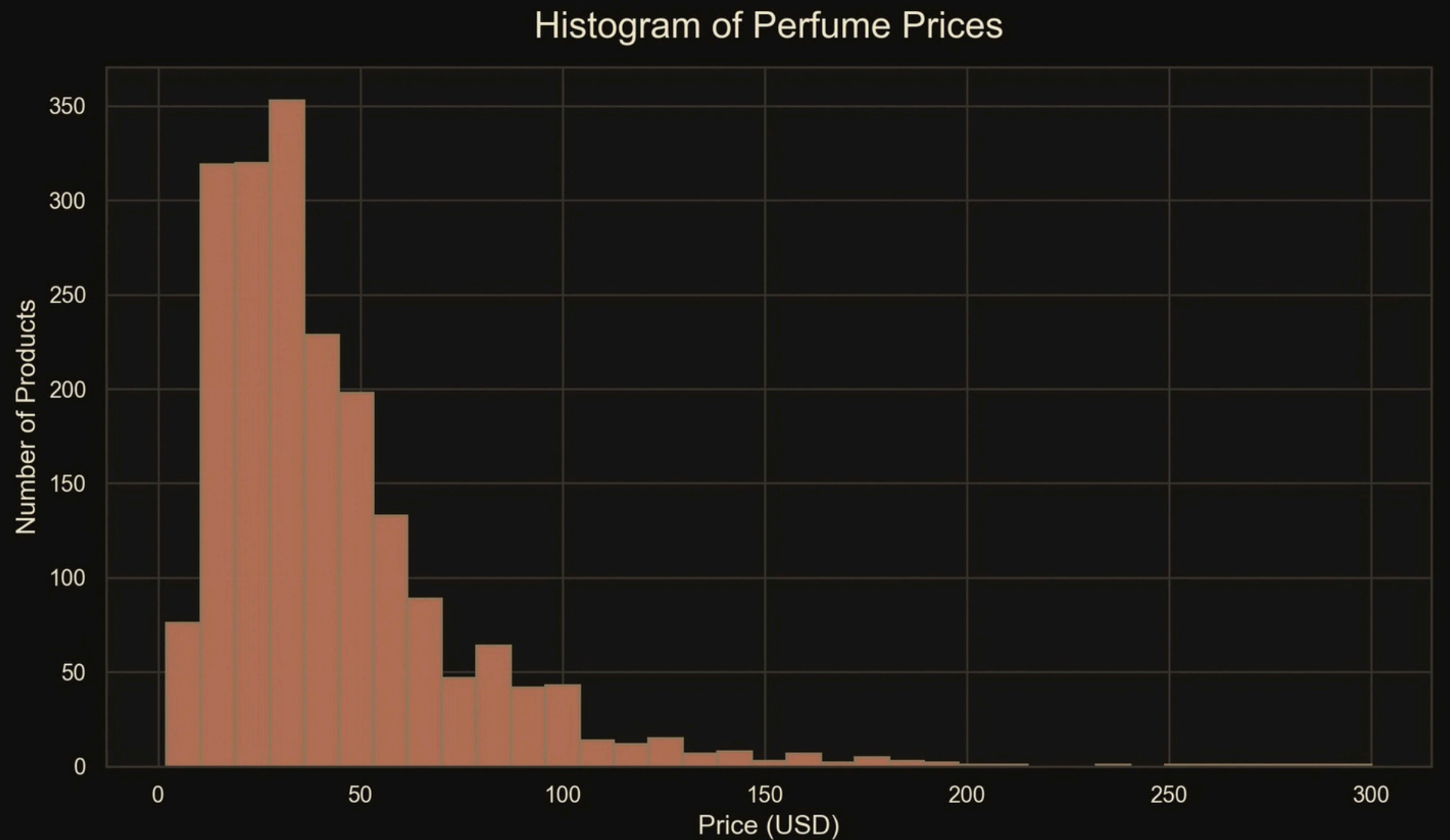


PRICE DISTRIBUTION ANALYSIS

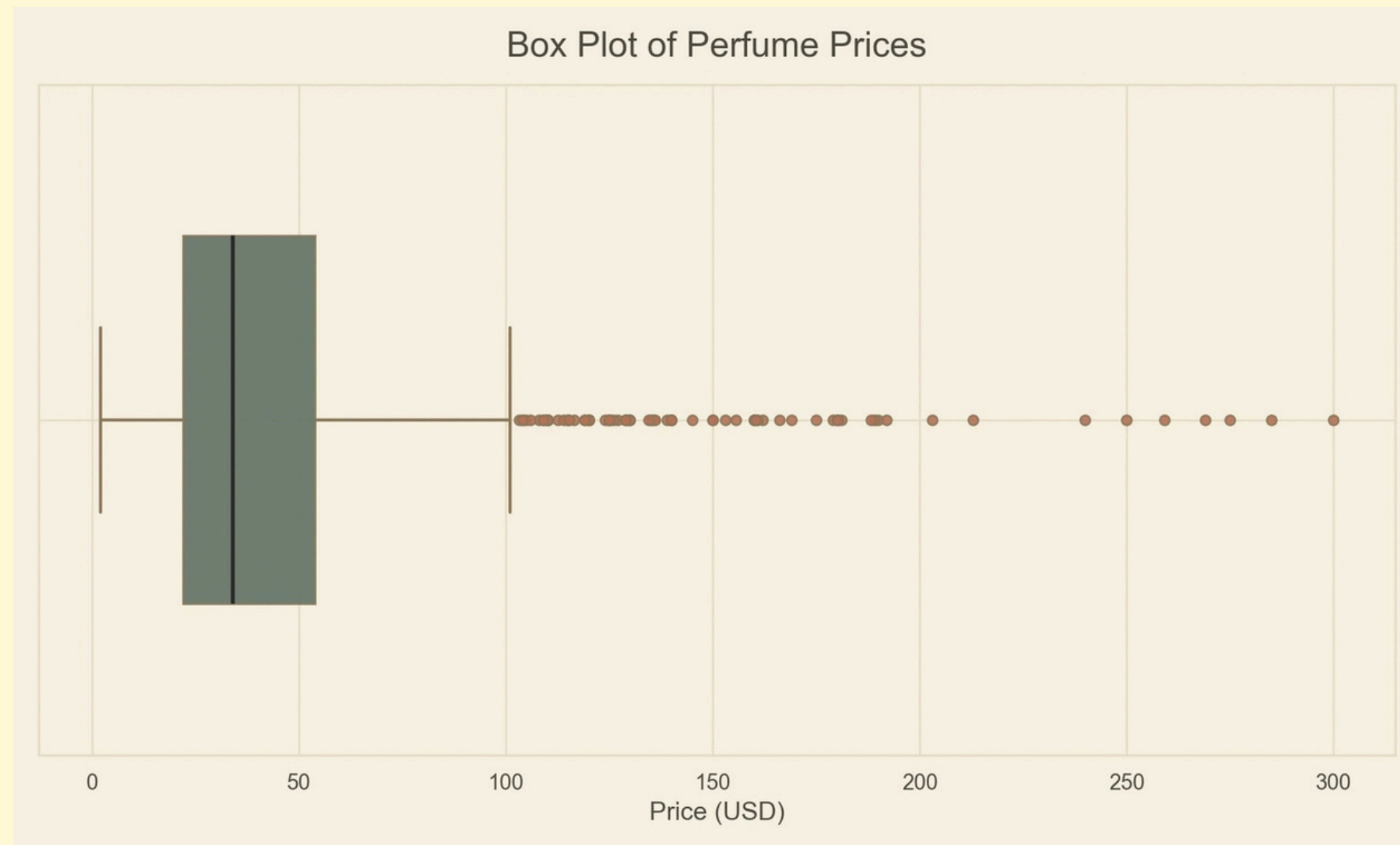
- **Technique:** Histogram / Univariate Numerical Analysis
- **Variable Type:** Univariate - 1 Numerical Feature
- **Features Used:** price

Key Insights:

- Low-to-mid price range.
- Strongly right-skewed.
- Number of premium products create a long upper tail.



OUTLIER DETECTION IN PRODUCT PRICES



- **Technique:** Box Plot / IQR-Based Outlier Analysis
- **Variable Type:** Univariate - 1 Numerical Feature
- **Features Used:** price

Key Insights:

- High-price outliers are clearly present in the dataset.
- Outliers are concentrated on the premium/luxury side.
- There is no strong low-price outlier issue.

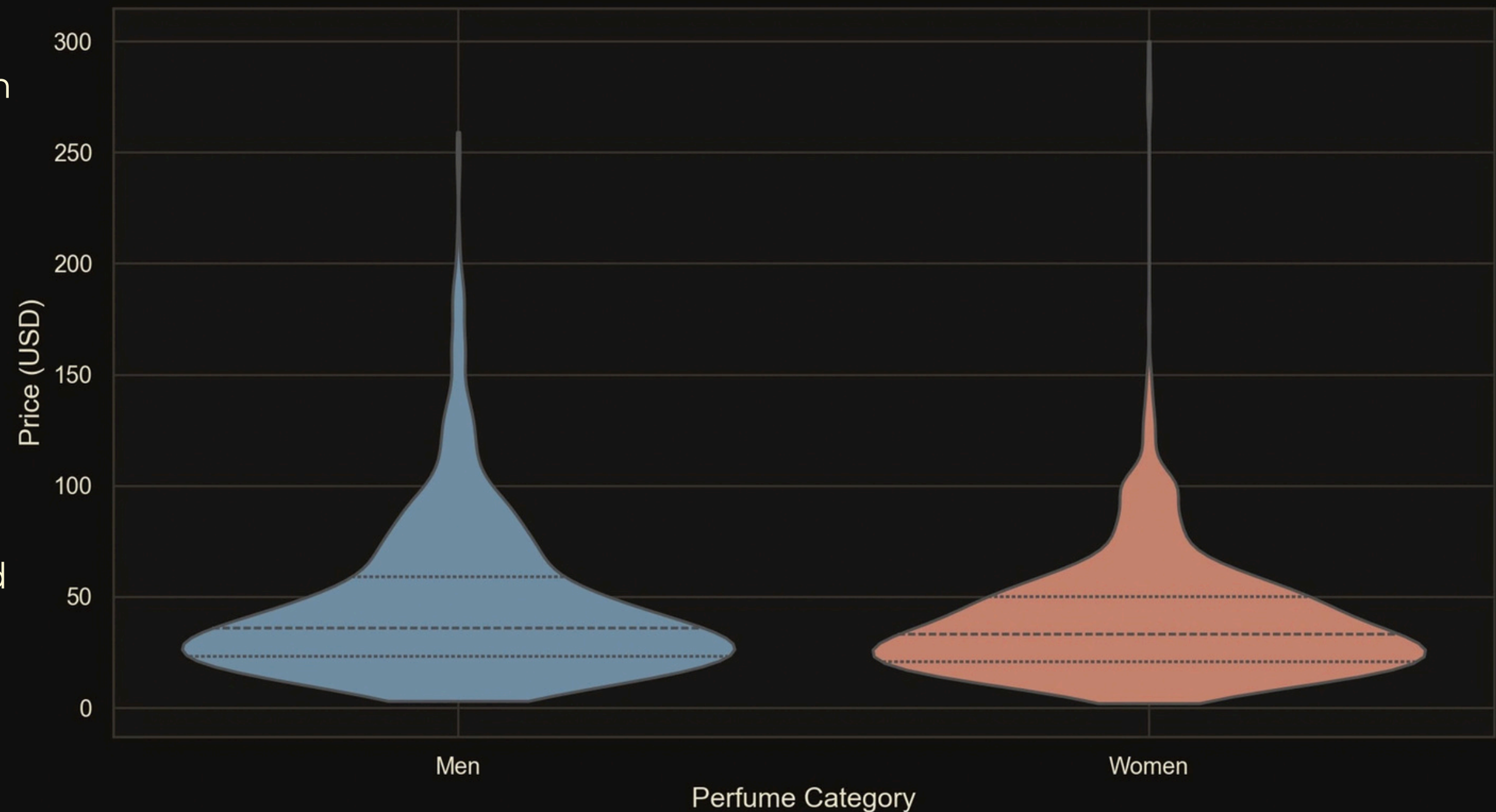
PRICE COMPARISON BY GENDER CATEGORY

- **Technique:** Violin Plot / Comparative Distribution Analysis
- **Variable Type:** Bivariate - 1 Numerical + 1 Categorical Feature
- **Features Used:** price (Numerical), gender (Categorical Grouping)
- **Target Type:** Binary Classification if modeled (Men / Women)

Key Insights:

- Men's perfumes have a slightly higher mean and median price.
- Both gender groups show right-skewed distributions.
- Premium products increase spread in both categories.

Violin Plot of Prices by Gender Category

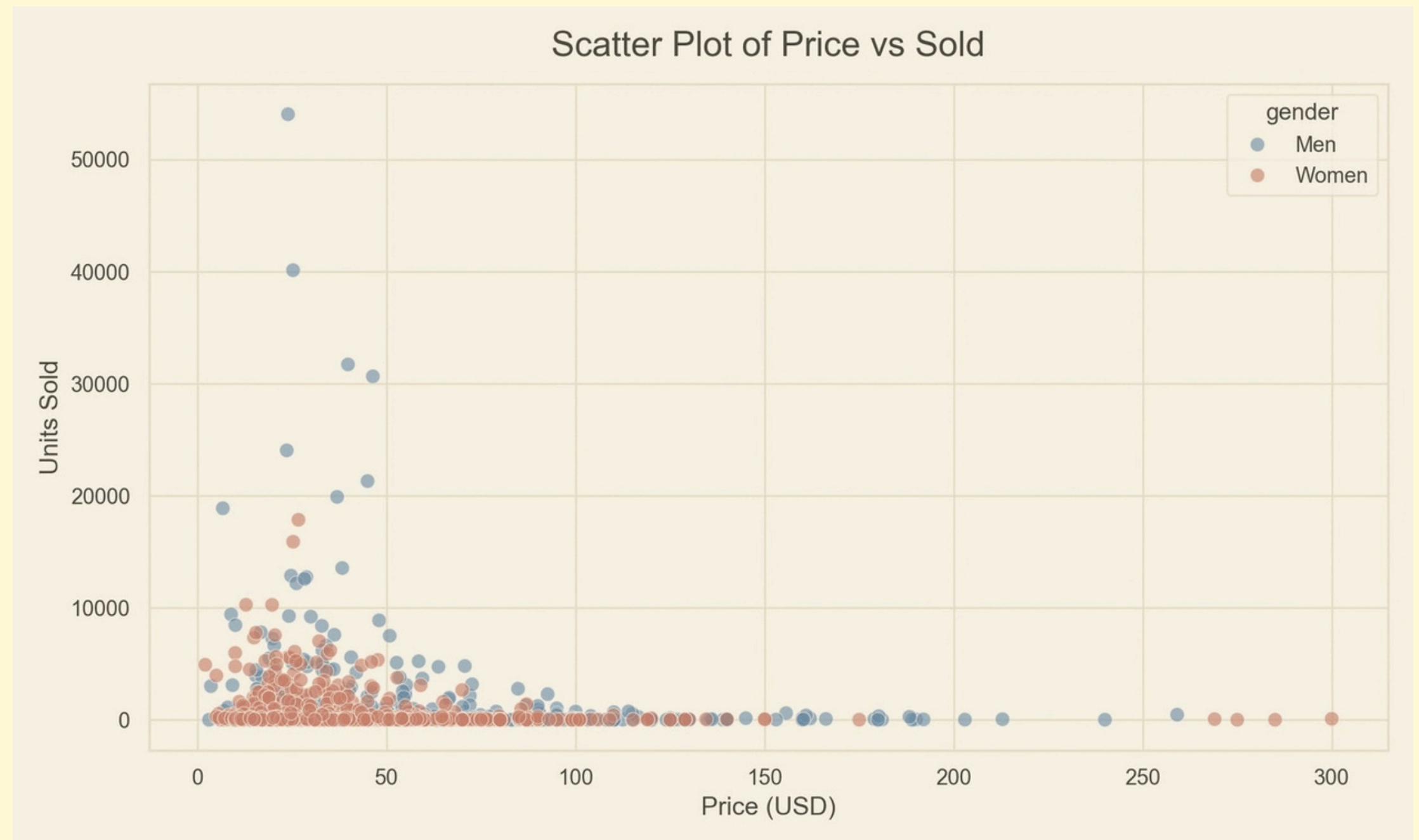


RELATIONSHIP BETWEEN PRICE AND OTHER NUMERICAL VARIABLES

- **Technique:** Scatter Plot
- **Variable Type:** Bivariate - 2 Numerical Features
- **Features Used:** price, sold
- **Target Type:** Regression if modeled (sold)

Key Insights:

- Price and sold have a weak negative relationship.
- Higher price does not necessarily lead to higher sales.
- Sales activity is more dispersed among low and mid-price products.



BRAND SEGMENTATION & GENDER FOCUS

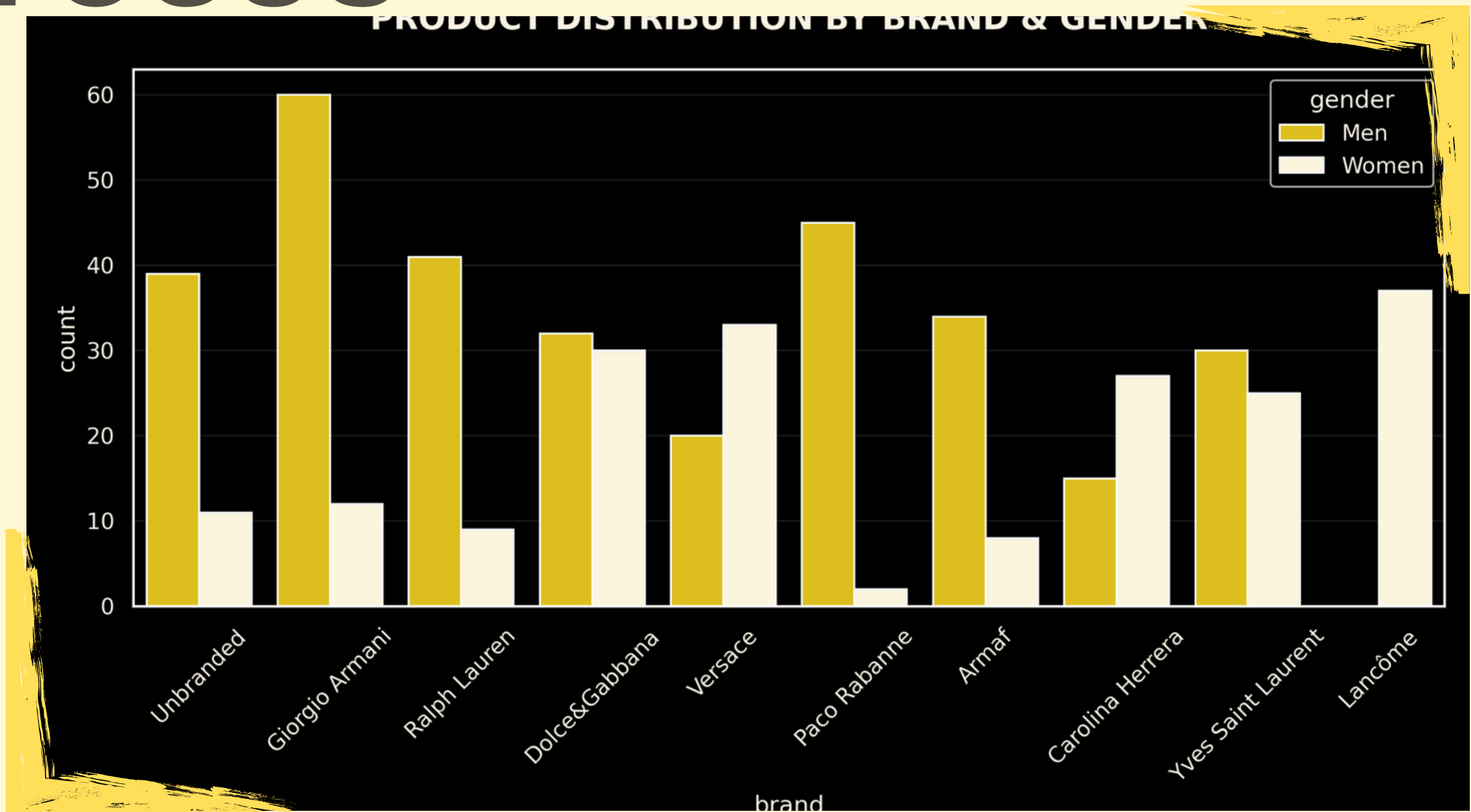


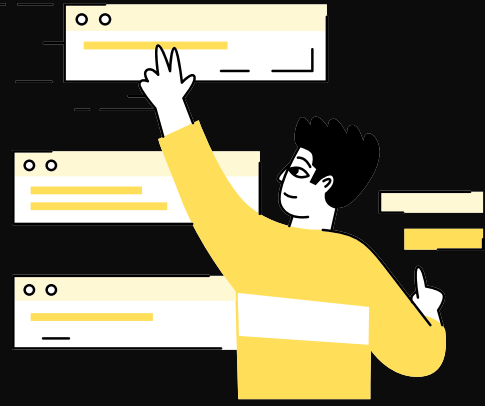
Technical Details

- **Technique:** Grouped Bar Chart (Categorical Analysis)
- **Variable Type:** Bivariate - 2 Categorical Features
- **Features Used:** brand (Independent), gender (Target/Color)
- **Target Type:** Binary Classification (Men / Women)

Key Insights:

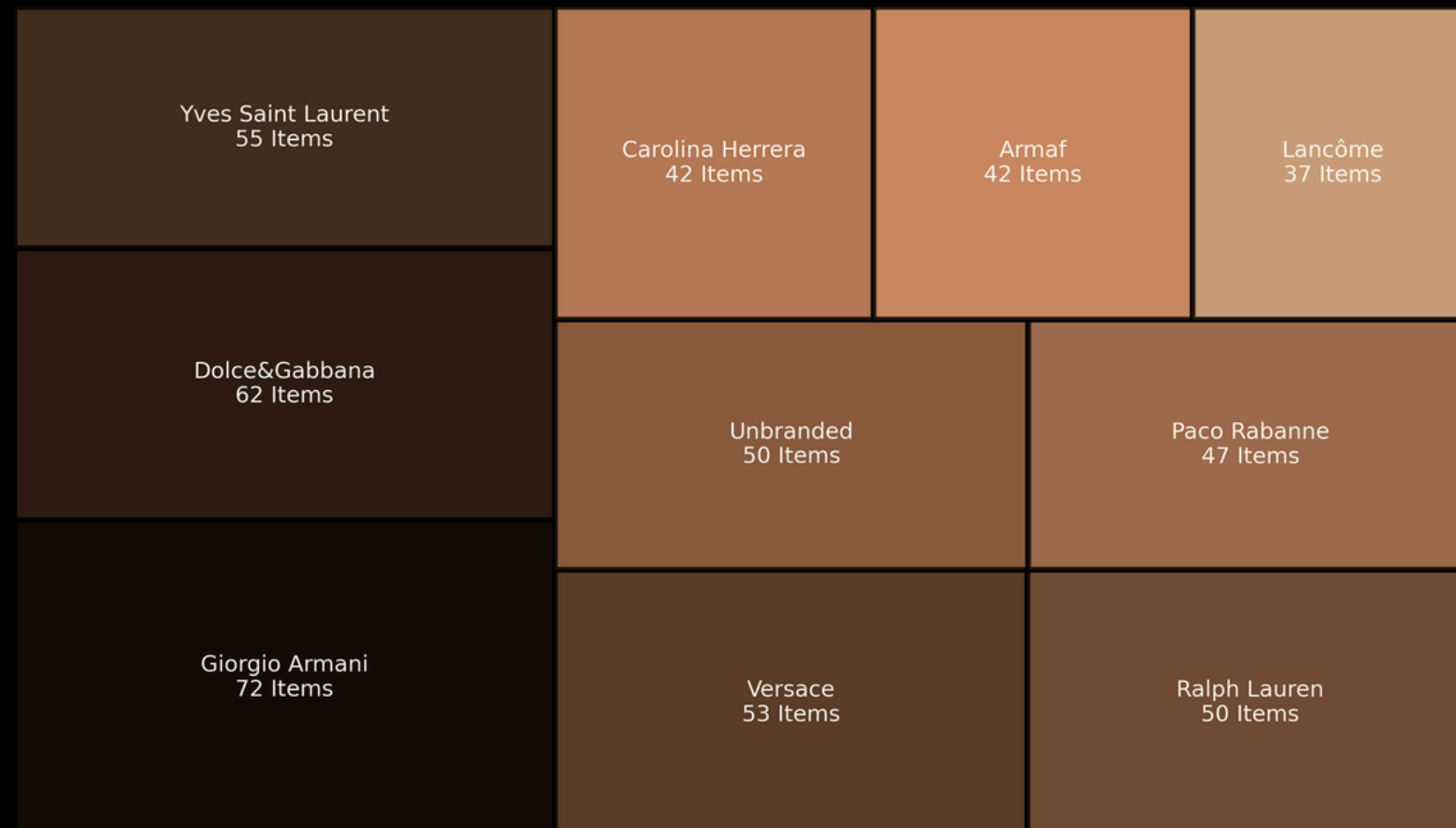
- **Brand Specialization**
- **Class Balance Check**
- **Market Dominance**





TOP 10 BRANDS & MARKET SHARE

TOP 10 BRANDS MARKET SHARE



Technical Details

- **Technique:** Treemap (Hierarchical Area Mapping)
- **Variable Type:** Univariate - Categorical Frequency Analysis
- **Features Used:** brand (1 Feature)

Key Insights

- Volume Distribution
- Competitive Landscape
- High Efficiency

PRICING STRATEGY BY BRAND

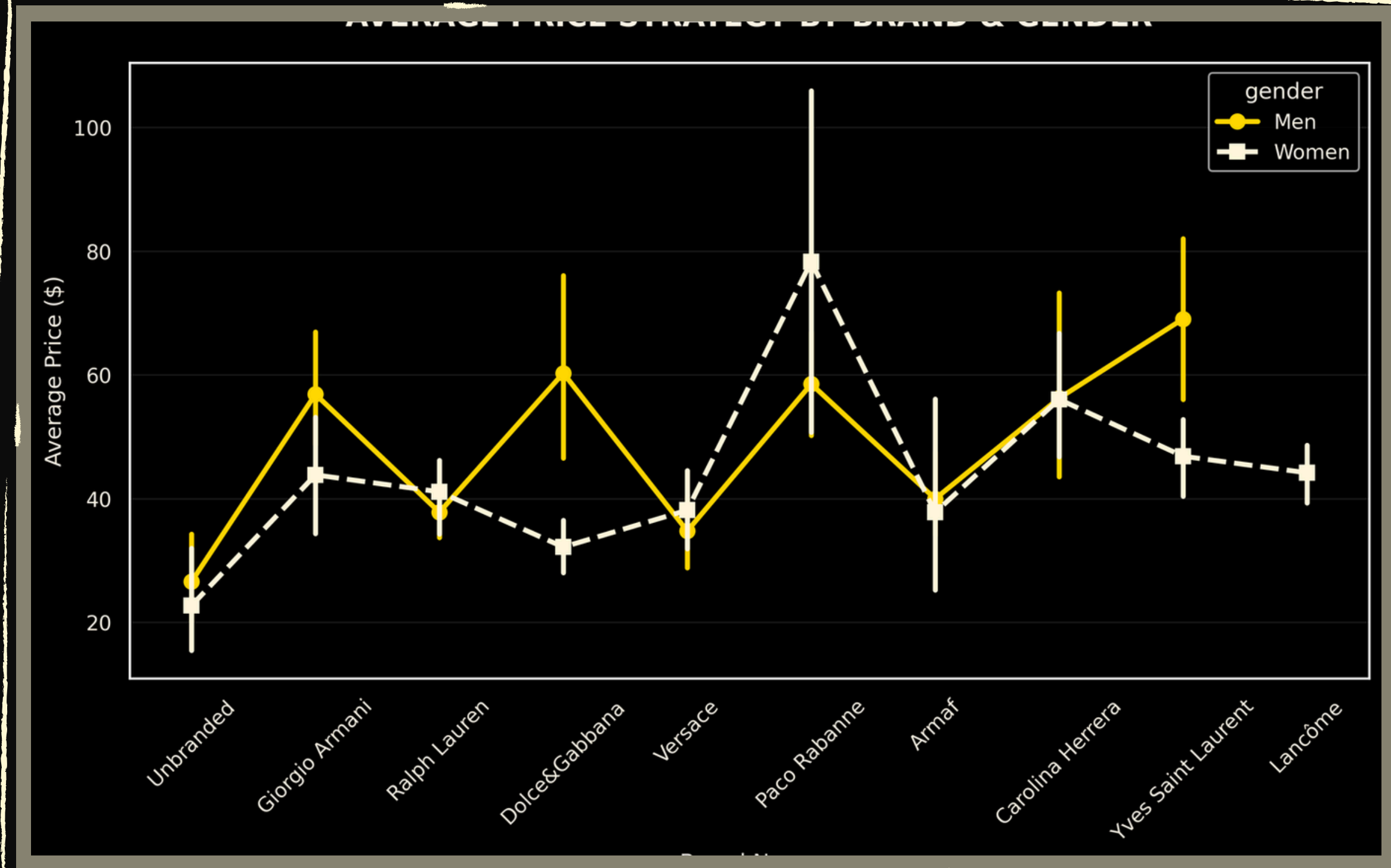


Technical Details

- **Technique:** Point Plot with Error Bars (Statistical Inference)
- **Variable Type:** Multivariate - 2 Categorical, 1 Numerical
- **Features Used:** brand , price , gender (3 features)

Key Insights

- **Strategic Positioning**
- **Price Stability**
- **Gender-Price Correlation**





CONCLUSION

Dataset Review Summary:

- **Repository:** Downloaded from Kaggle, containing two primary CSV files.
- **Complexity:** ~2000 samples with multiple categorical and numerical features.

Technical Summary:

- **Top Predictors:** brand and title show the highest Information Gain.
- **Relationship Type:** Complex, non-linear patterns detected.

Final Takeaways:

- **Segmentation:** Market is clearly split by brand volume and price points.
- **Business Value:** Insights provide a roadmap for e-commerce pricing and listing strategies.

